

Quarkyonic Matter Diego's Notes

Diego Juárez

Notes from Dr Vovchenko QM and group papers

May 20, 2026

Contents

Particle Number Fluctuations for van der Waals Equation of State	2
1.1 From Ideal gas to the van der Waals EOS	2
1.2 From CE to GCE	3
1.3 Redlich-Kwong-Soave (RKS) eos in the GCE	9
2 Van der Waals Equation of State with Fermi Statistics for Nuclear Matter	11
2.1 RKS Equation for the Boltzmann Statistics in the GCE	11
3 RKS Equation of State with Quantum Statistics	13
4 Quantum van der Waals theory meets quarkyonic matter	14
5 Quantum van der Waals quarkyonic matter at nonzero isospin asymmetry	15
6 Correlations between nuclear incompressibility, liquid-gas critical point, and quarkyonic transition	15
6.1 Equations of State of Real Gases	15
6.2 Ground state of nuclear matter	16
6.3 Computing critical points	21
6.3.1 Van der Waals example (classical)	21
6.3.2 8. Physics of the critical point	22
6.3.3 9. Mathematical procedure to compute T_c and n_c	22
6.3.4 10. Compact summary	23

Particle Number Fluctuations for van der Waals Equation of State

1.1 From Ideal gas to the van der Waals EOS

We begin with the ideal gas law,

$$PV = nRT. \quad (1)$$

This equation assumes:

- the particles have negligible volume,
- there are no intermolecular forces between particles.

These assumptions work well at low density, but they fail for real gases.

Therefore, to construct a more realistic equation of state, we introduce two corrections:

- i. a correction for the **finite size** of the particles,
- ii. a correction for the **attractive interactions** between particles.

i) Volume correction: finite molecular size

In the ideal gas model, particles are treated as point-like, so they can move in the full volume V .

For a real gas, however, molecules occupy space, so the volume actually available for motion is smaller than V .

If b is the excluded volume per mole, then for n moles the unavailable volume is approximately nb . Hence, the effective free volume becomes

$$V_{\text{free}} = V - nb. \quad (2)$$

Replacing V by $V - nb$ in the ideal gas expression gives

$$P \approx \frac{nRT}{V - nb}. \quad (3)$$

This is the first correction.

ii) Pressure correction: attractive interactions

In a real gas, molecules attract each other.

Because of this, a molecule approaching the wall is pulled slightly inward by neighboring molecules. As a consequence, the force exerted on the wall is reduced, and the observed pressure is smaller than it would be in the absence of attractions.

Thus, the pressure appearing in the ideal-like expression must be larger than the measured pressure P . We write

$$P_{\text{ideal-like}} = P + \Delta P. \quad (4)$$

Now we estimate ΔP .

The strength of the attractive correction should increase with the number of interacting pairs. Since the number density is

$$\rho = \frac{n}{V}, \quad (5)$$

the number of pair interactions scales as ρ^2 . Therefore,

$$\Delta P \propto \left(\frac{n}{V}\right)^2. \quad (6)$$

Introducing a proportionality constant a , which measures the strength of the attractive interactions, we obtain

$$\Delta P = a \left(\frac{n}{V}\right)^2 = a \frac{n^2}{V^2}. \quad (7)$$

Hence,

$$P_{\text{ideal-like}} = P + a \frac{n^2}{V^2}. \quad (8)$$

Combining both corrections

Now we replace:

$$\begin{aligned} \star \quad V &\rightarrow V - nb, \\ \star \quad P &\rightarrow P + a \frac{n^2}{V^2}. \end{aligned}$$

Then the ideal gas law becomes

$$\boxed{\left(P + a \frac{n^2}{V^2}\right) (V - nb) = nRT}, \quad (9)$$

this is the **van der Waals equation of state**, or:

$$\boxed{p(V, T, N) = \frac{NT}{V - bN} - a \frac{N^2}{V^2}}. \quad (10)$$

1.2 From CE to GCE

In the GCE the macroscopic states of a thermal system are defined in terms of the grand canonical potential Ω which is function of three physical variables (volume V , temperature T and chemical potential μ). It is connected to the system pressure as:

$$\Omega(V, T, \mu) = -p(T, \mu) V. \quad (11)$$

Therefore, pressure as a function of its natural variables, gives a complete description of the eos, one calculates the entropy S , (conserved) number density n , and energy density ϵ as:

$$S(T, \mu) = \left(\frac{\partial p}{\partial T}\right)_{\mu}, \quad n(T, \mu) = \left(\frac{\partial p}{\partial \mu}\right)_T, \quad \epsilon(T, \mu) = TS + \mu n - p. \quad (12)$$

The thermodynamical potential in the CE is the free energy $F(V, T, N)$. The function F , depends on its natural variables (T, V, N) and it gives a complete thermodynamical description of the system properties. Entropy, pressure, chemical potential and the total energy can be calculated in the CE as:

$$S(V, T, N) = -\left(\frac{\partial F}{\partial T}\right)_{V, N}, \quad p(V, T, N) = -\left(\frac{\partial F}{\partial V}\right)_{T, N}, \quad \mu(V, T, N) = \left(\frac{\partial F}{\partial N}\right)_{T, V} \quad (13)$$

and

$$F + TS \quad (14)$$

If one knows pressure in the CE it is not yet possible to find entropy, chemical potential and energy in unique way. For a complete thermodynamical description in CE one needs to reconstruct free energy, thus, the following should be solved:

$$\left(\frac{\partial F}{\partial V} = -p(V, T, N) \right). \quad (15)$$

The solution of later eq can derived as:

$$F(V, T, N) = F(V_0, T, N) - \int_{V_0}^V p(V', T, N) dV \quad (16)$$

To find $F(V, T, N)$ one needs to know not only pressure but, additionally, the free energy $F(V_0, T, N)$ as the function of T and N at some fixed value $V = V_0$. We assume that for fixed T the role of particle interactions becomes negligible at very small density $N/V_0 \rightarrow 0$. For relativistic ideal Boltzmann gas the free energy reads

$$F_{\text{id}}(V, T, N) = -NT \left[1 + \ln \left(\frac{V \phi(T; d, m)}{N} \right) \right] \quad (17)$$

with

$$\phi(T; d, m) = \frac{d}{2\pi^2} \int_0^\infty k^2 dk \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right] = \frac{d m^2 T}{2\pi^2} K_2 \left(\frac{m}{T} \right). \quad (18)$$

Full derivation from later eq:

We begin from the canonical ensemble relation

$$F_{\text{id}}(V, T, N) = -T \ln Z(V, T, N), \quad (19)$$

where $Z(V, T, N)$ is the canonical partition function.

For a gas of N identical, non-interacting classical particles,

$$Z(V, T, N) = \frac{1}{N!} [Z_1(V, T)]^N, \quad (20)$$

where $Z_1(V, T)$ is the one-particle partition function.

For a relativistic particle with energy

$$\varepsilon(k) = \sqrt{k^2 + m^2}, \quad (21)$$

the one-particle partition function is

$$Z_1(V, T) = d \int \frac{d^3x d^3k}{(2\pi)^3} \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (22)$$

Here, d is the degeneracy factor. Since the integrand does not depend on position,

$$\int d^3x = V, \quad (23)$$

thus

$$Z_1(V, T) = V d \int \frac{d^3k}{(2\pi)^3} \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (24)$$

Now define

$$\phi(T; d, m) = d \int \frac{d^3 k}{(2\pi)^3} \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (25)$$

Then

$$Z_1(V, T) = V \phi(T; d, m), \quad (26)$$

and therefore

$$Z(V, T, N) = \frac{[V \phi(T; d, m)]^N}{N!}. \quad (27)$$

Using

$$F_{\text{id}}(V, T, N) = -T \ln Z(V, T, N), \quad (28)$$

we get

$$F_{\text{id}}(V, T, N) = -T \ln \left(\frac{[V \phi(T; d, m)]^N}{N!} \right) \quad (29)$$

$$= -T [N \ln(V \phi(T; d, m)) - \ln N!]. \quad (30)$$

For large N , we use Stirling's approximation,

$$\ln N! \approx N \ln N - N. \quad (31)$$

Substituting this into the free energy expression,

$$F_{\text{id}}(V, T, N) = -T [N \ln(V \phi(T; d, m)) - N \ln N + N] \quad (32)$$

$$= -NT [\ln(V \phi(T; d, m)) - \ln N + 1]. \quad (33)$$

Using

$$\ln A - \ln B = \ln \left(\frac{A}{B} \right), \quad (34)$$

we obtain

$$F_{\text{id}}(V, T, N) = -NT \left[1 + \ln \left(\frac{V \phi(T; d, m)}{N} \right) \right]. \quad (35)$$

We now evaluate

$$\phi(T; d, m) = d \int \frac{d^3 k}{(2\pi)^3} \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (36)$$

Since the integrand depends only on the magnitude $k = |\mathbf{k}|$, we use spherical coordinates in momentum space:

$$d^3 k = 4\pi k^2 dk. \quad (37)$$

Thus,

$$\phi(T; d, m) = d \frac{4\pi}{(2\pi)^3} \int_0^\infty k^2 dk \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right] \quad (38)$$

$$= \frac{d}{2\pi^2} \int_0^\infty k^2 dk \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (39)$$

Hence,

$$\phi(T; d, m) = \frac{d}{2\pi^2} \int_0^\infty k^2 dk \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (40)$$

Consider the integral

$$I = \int_0^\infty k^2 dk \exp \left[-\frac{\sqrt{k^2 + m^2}}{T} \right]. \quad (41)$$

Introduce the change of variables

$$k = m \sinh \theta, \quad \sqrt{k^2 + m^2} = m \cosh \theta, \quad dk = m \cosh \theta d\theta. \quad (42)$$

Then

$$k^2 dk = m^2 \sinh^2 \theta (m \cosh \theta d\theta) = m^3 \sinh^2 \theta \cosh \theta d\theta. \quad (43)$$

Therefore,

$$I = m^3 \int_0^\infty \sinh^2 \theta \cosh \theta e^{-(m/T) \cosh \theta} d\theta. \quad (44)$$

Now use

$$\sinh^2 \theta = \cosh^2 \theta - 1, \quad (45)$$

so that

$$\sinh^2 \theta \cosh \theta = \cosh^3 \theta - \cosh \theta. \quad (46)$$

Thus,

$$I = m^3 \int_0^\infty (\cosh^3 \theta - \cosh \theta) e^{-(m/T) \cosh \theta} d\theta. \quad (47)$$

Using the identity

$$\cosh 3\theta = 4 \cosh^3 \theta - 3 \cosh \theta, \quad (48)$$

we solve for $\cosh^3 \theta$:

$$\cosh^3 \theta = \frac{1}{4} (\cosh 3\theta + 3 \cosh \theta). \quad (49)$$

Hence,

$$\cosh^3 \theta - \cosh \theta = \frac{1}{4} (\cosh 3\theta + 3 \cosh \theta) - \cosh \theta \quad (50)$$

$$= \frac{1}{4} (\cosh 3\theta - \cosh \theta). \quad (51)$$

Therefore,

$$I = \frac{m^3}{4} \int_0^\infty (\cosh 3\theta - \cosh \theta) e^{-(m/T) \cosh \theta} d\theta. \quad (52)$$

Now use the integral representation of the modified Bessel function of the second kind,

$$K_\nu(z) = \int_0^\infty e^{-z \cosh t} \cosh(\nu t) dt. \quad (53)$$

Then

$$I = \frac{m^3}{4} \left[K_3\left(\frac{m}{T}\right) - K_1\left(\frac{m}{T}\right) \right]. \quad (54)$$

Next, apply the recurrence relation

$$K_{\nu+1}(z) - K_{\nu-1}(z) = \frac{2\nu}{z} K_\nu(z). \quad (55)$$

Setting $\nu = 2$, we obtain

$$K_3(z) - K_1(z) = \frac{4}{z} K_2(z). \quad (56)$$

With $z = m/T$, this gives

$$I = \frac{m^3}{4} \cdot \frac{4T}{m} K_2\left(\frac{m}{T}\right) = m^2 T K_2\left(\frac{m}{T}\right). \quad (57)$$

Thus,

$$\int_0^\infty k^2 dk \exp\left[-\frac{\sqrt{k^2 + m^2}}{T}\right] = m^2 T K_2\left(\frac{m}{T}\right). \quad (58)$$

Substituting into $\phi(T; d, m)$,

$$\phi(T; d, m) = \frac{d}{2\pi^2} m^2 T K_2\left(\frac{m}{T}\right). \quad (59)$$

Hence,

$$\phi(T; d, m) = \frac{d m^2 T}{2\pi^2} K_2\left(\frac{m}{T}\right). \quad (60)$$

Combining everything,

$$F_{\text{id}}(V, T, N) = -NT \left[1 + \ln\left(\frac{V \phi(T; d, m)}{N}\right) \right] \quad (61)$$

with

$$\phi(T; d, m) = \frac{d}{2\pi^2} \int_0^\infty k^2 dk \exp\left[-\frac{\sqrt{k^2 + m^2}}{T}\right] = \frac{d m^2 T}{2\pi^2} K_2\left(\frac{m}{T}\right). \quad (62)$$

In eqs 17 and 18, d is the degeneracy factor, m is the particle mass and quantity ϕ has a physical meaning of the GCE particle number density at zero chemical potential.

Substituting in eq 16

$$F(V, T, N) = -NT \left[1 + \ln \frac{V_0 \phi(T; d, m)}{N} \right] + \int_V^{V_0} \frac{NT}{V' - bN} \quad (63)$$

$$F(V, T, N) = -NT \left[1 + \ln \frac{V_0 \phi(T; d, m)}{N} \right] + NT \int_V^{V_0} \frac{1}{V' - bN} \quad (64)$$

let be $u = V' - bN$, the limits become

$$V_0 - bN = u_0$$

and

$$F(V, T, N) = -NT \left[1 + \ln \frac{V_0 \phi(T; d, m)}{N} \right] + NT \int_u^{u_0} \frac{1}{u} \quad (65)$$

$$F(V, T, N) = -NT \left[1 + \ln \frac{V_0 \phi(T; d, m)}{N} \right] + NT \ln u \Big|_{V-bN}^{V_0-bN} \quad (66)$$

$$F(V, T, N) = -NT \left[1 + \ln \frac{V_0 \phi(T; d, m)}{N} \right] + NT \ln \frac{V_0 - bN}{V - bN} = F_{\text{id}}(V - bN, T, N) \quad (67)$$

For the free energy (later eq), the chemical potential can be calculated as:

$$F(V, T, N) = -NT \left[1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right] + NT \ln \left(\frac{V_0 - bN}{V - bN} \right) = F_{\text{id}}(V - bN, T, N) \quad (68)$$

For the free energy above, the chemical potential is

$$\mu = \left(\frac{\partial F}{\partial N} \right)_{V,T}.$$

Starting from

$$F = -NT \left[1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right] + NT \ln \left(\frac{V_0 - bN}{V - bN} \right),$$

we differentiate term by term:

$$\mu = \frac{\partial}{\partial N} \left[-NT \left(1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right) \right] + \frac{\partial}{\partial N} \left[NT \ln \left(\frac{V_0 - bN}{V - bN} \right) \right]. \quad (69)$$

For the first term,

$$\frac{\partial}{\partial N} \left[-NT \left(1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right) \right] = -T \left(1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right) - NT \frac{\partial}{\partial N} \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \quad (70)$$

$$= -T \left(1 + \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) \right) - NT \left(-\frac{1}{N} \right) \quad (71)$$

$$= -T \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right). \quad (72)$$

For the second term,

$$\frac{\partial}{\partial N} \left[NT \ln \left(\frac{V_0 - bN}{V - bN} \right) \right] = T \ln \left(\frac{V_0 - bN}{V - bN} \right) + NT \frac{\partial}{\partial N} \ln \left(\frac{V_0 - bN}{V - bN} \right) \quad (73)$$

$$= T \ln \left(\frac{V_0 - bN}{V - bN} \right) + NT \left[\frac{-b}{V_0 - bN} - \left(\frac{-b}{V - bN} \right) \right] \quad (74)$$

$$= T \ln \left(\frac{V_0 - bN}{V - bN} \right) + bNT \left(\frac{1}{V - bN} - \frac{1}{V_0 - bN} \right). \quad (75)$$

Therefore,

$$\mu = -T \ln \left(\frac{V_0 \phi(T; d, m)}{N} \right) + T \ln \left(\frac{V_0 - bN}{V - bN} \right) + bNT \left(\frac{1}{V - bN} - \frac{1}{V_0 - bN} \right). \quad (76)$$

where we have used that $(V_0 - bN)/V_0 \rightarrow 1$ at $V_0 \rightarrow \infty$

$$\mu = -T \ln \left(\frac{(V - bN) \phi(T; d, m)}{N} \right) + \frac{bNT}{V - bN}. \quad (77)$$

For particle number density $n = N/V$ one finds that for the ideal gas, set $b = 0$ in Eq. (77). Then

$$\mu = -T \ln \left(\frac{V \phi(T; d, m)}{N} \right). \quad (78)$$

Hence

$$\frac{\mu}{T} = -\ln \left(\frac{V \phi(T; d, m)}{N} \right), \quad (79)$$

and exponentiating both sides gives

$$\exp \left(\frac{\mu}{T} \right) = \frac{N}{V \phi(T; d, m)}. \quad (80)$$

Therefore,

$$\frac{N}{V} = \exp\left(\frac{\mu}{T}\right) \phi(T; d, m) \equiv n_{\text{id}}(T, \mu). \quad (81)$$

On the other hand at $b > 0$ easily we have:

$$\frac{N}{V} \equiv n_{ev}(T, V) = \frac{n_{id}(T, \mu^*)}{1 + bn_{id}(T, \mu^*)}, \quad \mu^* = \mu - b \frac{n_{ev}(T, \mu)T}{1 - bn_{ev}(T, \mu)}, \quad (82)$$

or equivalently

$$p_{ev}(T, V) = p_{id}(T, \mu), \quad \mu^* = \mu - b \frac{n_{ev}(T, \mu)T}{1 - bn_{ev}(T, \mu)} = \mu - bp_{ev}(T, \mu) \quad (83)$$

Therefore, to solve the excluded-volume problem in case of the Boltzmann statistics one has to obtain solution of transcendental equation 82 with respect to particle density $n_{ev}(T, \mu)$ or of trascendental eq 83 with respect to pressure $p_{ev}(T, \mu)$ for every point in (μ, T) -plane, i.e. in the GCE.

1.3 Redlich-Kwong-Soave (RKS) eos in the GCE

In the grand canonical the rks eq for the system pressure is defined as:

$$P_{\text{RKS}}(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + bn}, \quad (84)$$

or

$$P_{\text{RKS}}(T, V, N) = \frac{\frac{N}{V}T}{1 - b\frac{N}{V}} - \frac{a(\frac{N}{V})^2}{1 + b\frac{N}{V}}, \quad (85)$$

$$P_{\text{RKS}}(T, N, V) = \frac{NT}{V - bN} - \frac{aN^2}{V(V + bN)}. \quad (86)$$

So using eq 16:

$$\begin{aligned} F(V, T, N) &= F_{\text{id}}(V_0, T, N) + \int_V^{V_0} dV' \left[\frac{NT}{V' - bN} - \frac{aN^2}{V'(V' + bN)} \right] \\ &\simeq F_{\text{id}}(V - bN, T, N) + \frac{aN}{b} \ln \left(\frac{V}{V + bN} \right). \end{aligned} \quad (87)$$

The chemical potential then reads

$$\begin{aligned} \mu &= \left(\frac{\partial F}{\partial N} \right)_{V, T} \\ &= -T \ln \left[\frac{(V - bN)\phi(T; d, m)}{N} \right] + \frac{bNT}{V - bN} + \frac{a}{b} \ln \left(\frac{V}{V + bN} \right) - \frac{aN}{V + bN}. \end{aligned} \quad (88)$$

Introducing the particle density

$$n \equiv \frac{N}{V}, \quad (89)$$

Eq. (88) can be written as

$$\mu = -T \ln \left[\frac{(1 - bn)\phi(T; d, m)}{n} \right] + \frac{bnT}{1 - bn} - \frac{a}{b} \ln(1 + bn) - \frac{an}{1 + bn}. \quad (90)$$

Therefore, the GCE particle density is obtained from the transcendental equation

$$n(T, \mu) = \frac{n_{\text{id}}(T, \mu^*)}{1 + b n_{\text{id}}(T, \mu^*)}, \quad (91)$$

where

$$\mu^* = \mu - \frac{bnT}{1 - bn} + \frac{a}{b} \ln(1 + bn) + \frac{an}{1 + bn}. \quad (92)$$

The function $n(T, \mu)$, which is the solution of the transcendental equation (91), is reduced to the excluded-volume result if $a = 0$, and to the ideal-gas result if $a = b = 0$. The parameter $b > 0$ describes the short-range repulsion between particles and suppresses the particle number density, while the attractive interaction controlled by $a > 0$ enhances the density.

The energy density of the RKS gas can be calculated within the GCE from the pressure function, $p(T, \mu)$,

$$\begin{aligned} \epsilon(T, \mu) &= T \left(\frac{\partial p}{\partial T} \right)_{\mu} + \mu \left(\frac{\partial p}{\partial \mu} \right)_T - p \\ &= \frac{\epsilon_{\text{id}}(T, \mu^*)}{1 + b n_{\text{id}}(T, \mu^*)} - \frac{an}{b} \ln(1 + bn) \\ &= \left[\bar{\epsilon}_{\text{id}}(T; m) - \frac{a}{b} \ln(1 + bn) \right] n, \end{aligned} \quad (93)$$

where μ^* is defined as

$$\mu^* = \mu - \frac{bnT}{1 - bn} + \frac{a}{b} \ln(1 + bn) + \frac{an}{1 + bn}, \quad (94)$$

and

$$\bar{\epsilon}_{\text{id}}(T; m) = \frac{\int_0^\infty k^2 dk \sqrt{k^2 + m^2} \exp \left[-\sqrt{k^2 + m^2}/T \right]}{\int_0^\infty k^2 dk \exp \left[-\sqrt{k^2 + m^2}/T \right]} = 3T + m \frac{K_1(m/T)}{K_2(m/T)} \quad (95)$$

is the average energy per particle in the ideal gas.

The repulsive parameter $b > 0$ suppresses the particle number density by transforming n_{id} into n . The attractive parameter $a > 0$ changes the average energy per particle from $\bar{\epsilon}_{\text{id}}$ to

$$\bar{\epsilon}_{\text{RKS}} = \bar{\epsilon}_{\text{id}}(T; m) - \frac{a}{b} \ln(1 + bn). \quad (96)$$

In the limit $b \rightarrow 0$, one recovers the VDW result,

$$\frac{a}{b} \ln(1 + bn) \rightarrow an, \quad \bar{\epsilon}_{\text{RKS}} \rightarrow \bar{\epsilon}_{\text{id}} - an. \quad (97)$$

It is also instructive to consider another derivation of the GCE equation, based on the procedure used in Ref. [?]. Introducing the particle number density,

$$n = \frac{N}{V}, \quad (98)$$

the RKS pressure function can be written as

$$p(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + bn}. \quad (99)$$

For the particle number density in the GCE, the following differential equation can be obtained:

$$\begin{aligned} n(T, \mu) &= \left(\frac{\partial p(T, \mu)}{\partial \mu} \right)_T \\ &= \left(\frac{\partial p(T, n)}{\partial n} \right)_T \left(\frac{\partial n(T, \mu)}{\partial \mu} \right)_T \\ &= \left[\frac{T}{(1 - bn)^2} - \frac{an(2 + bn)}{(1 + bn)^2} \right] \left(\frac{\partial n(T, \mu)}{\partial \mu} \right)_T. \end{aligned} \quad (100)$$

A general solution of Eq. (100) can be rewritten as

$$\begin{aligned}\mu &= \int dn \left[\frac{T}{n(1-bn)^2} - \frac{a(2+bn)}{(1+bn)^2} \right] \\ &= T \ln \left[\frac{n}{\phi(T; d, m)(1-bn)} \right] + \frac{bTn}{1-bn} - \frac{a}{b} \ln(1+bn) - \frac{an}{1+bn}.\end{aligned}\quad (101)$$

Therefore, the temperature-dependent shifted chemical potential is defined as

$$\mu^* = \mu - \frac{bTn}{1-bn} + \frac{a}{b} \ln(1+bn) + \frac{an}{1+bn}.\quad (102)$$

The particle density in the GCE is then obtained from the transcendental equation

$$n(T, \mu) = \frac{n_{\text{id}}(T, \mu^*)}{1 + b n_{\text{id}}(T, \mu^*)}.\quad (103)$$

2 Van der Waals Equation of State with Fermi Statistics for Nuclear Matter

Although this paper treats vdw model with quantum statistics i will described it with the rks model to understand better the procedure made by Dr Vovchenko.

2.1 RKS Equation for the Boltzmann Statistics in the GCE

The RKS pressure function corresponds to the Boltzmann approximation, i.e., the effects of quantum statistics are neglected. In the last section the RKS equation of state was formulated in the GCE. The GCE pressure, $p(T, \mu)$, is a function of temperature T and chemical potential μ . It contains complete information about thermodynamic functions of the system. Particle number density $n(T, \mu)$, entropy density $s(T, \mu)$, and energy density $\epsilon(T, \mu)$ can be presented in terms of p and its derivatives:

$$n(T, \mu) = \left(\frac{\partial p}{\partial \mu} \right)_T, \quad s(T, \mu) = \left(\frac{\partial p}{\partial T} \right)_\mu, \quad \epsilon(T, \mu) = T \left(\frac{\partial p}{\partial T} \right)_\mu + \mu \left(\frac{\partial p}{\partial \mu} \right)_T - p.\quad (104)$$

For $a = b = 0$, the above RKS equations reduce to the ideal-gas expressions for classical particles.

The RKS equation of state in the GCE is obtained in the form of a transcendental equation for the particle number density $n \equiv n(T, \mu)$, as a function of T and μ :

$$n(T, \mu) = \frac{n_{\text{id}}(T, \mu^*)}{1 + b n_{\text{id}}(T, \mu^*)}, \quad \mu^* = \mu - \frac{bnT}{1-bn} + \frac{a}{b} \ln(1+bn) + \frac{an}{1+bn}.\quad (105)$$

Here n^{id} is the particle number density in the ideal Boltzmann gas,

$$n^{\text{id}}(T, \mu) = \exp \left(\frac{\mu}{T} \right) \frac{dm^2 T}{2\pi^2} K_2 \left(\frac{m}{T} \right),\quad (106)$$

where d is the degeneracy factor, m is the particle mass, and $K_2(x)$ is the modified Bessel function of the second kind. The relativistic dispersion relation is considered,

$$\omega(k) = \sqrt{m^2 + k^2}.\quad (107)$$

The shifted chemical potential μ^* contains the effects of both the repulsive and attractive interactions. The parameter $b > 0$ describes the repulsive interaction and suppresses the particle number density through

the excluded-volume denominator in Eq. (105). The parameter $a > 0$ describes the attractive interaction and modifies the chemical potential through the logarithmic term and the rational term in Eq. (105). In the limit $a = 0$, the RKS equation reduces to the excluded-volume equation of state, while for $a = b = 0$, it reduces to the ideal Boltzmann gas.

The RKS pressure $p(T, n)$ is a unique function of variables T and n for all $T \geq 0$ and $0 \leq n < 1/b$. The RKS equation of state contains a first-order liquid-gas phase transition and has a critical point. The critical point (T_c, n_c) corresponds to the temperature and particle number density where

$$\left(\frac{\partial p}{\partial n}\right)_T = 0, \quad \left(\frac{\partial^2 p}{\partial n^2}\right)_T = 0. \quad (108)$$

For the RKS pressure

$$p(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + bn}, \quad (109)$$

one obtains

$$\left(\frac{\partial p}{\partial n}\right)_T = \frac{T}{(1 - bn)^2} - \frac{an(2 + bn)}{(1 + bn)^2}, \quad (110)$$

and

$$\left(\frac{\partial^2 p}{\partial n^2}\right)_T = \frac{2bT}{(1 - bn)^3} - \frac{2a}{(1 + bn)^3}. \quad (111)$$

Solving Eqs. (110) and (111) simultaneously gives the critical quantities

$$n_c = \frac{2^{1/3} - 1}{b}, \quad T_c = \frac{a}{2b} \left(2 - 2^{1/3}\right)^3, \quad p_c = \frac{a}{b^2} \left(1 + 3 \cdot 2^{1/3} - 3 \cdot 2^{2/3}\right). \quad (112)$$

At $T > T_c$, the following equation is always valid:

$$\left(\frac{\partial p}{\partial n}\right)_T > 0. \quad (113)$$

At $T < T_c$, the unstable region appears where

$$\left(\frac{\partial p}{\partial n}\right)_T < 0. \quad (114)$$

Therefore, the RKS isotherm $p(n, T)$ at $T < T_c$ has a local maximum and a local minimum. The unstable part of the RKS isotherm must be replaced by the Maxwell construction, which transforms the system into a mixture of two phases: a gas with density n_g and a liquid with density n_l , with $n_g < n_l$.

In the GCE, the mixed phase appears in a different way. At $T > T_c$, there is a unique solution of the transcendental equation for the particle number density $n(T, \mu)$. At $T < T_c$, one or three different solutions may appear for the RKS pressure $p(T, \mu)$. If several solutions are possible, the physical solution is the one with the largest pressure, in accordance with the Gibbs criterion.

The classical Boltzmann statistics leads to nonphysical behavior in the zero-temperature limit. This is already seen at the ideal-gas level. For the ideal Boltzmann gas, the entropy density in the nonrelativistic limit $T/m \ll 1$ is

$$s_{\text{Boltz}}^{\text{id}} = n^{\text{id}} \left[\frac{m}{T} + \frac{5}{2} - \frac{\mu}{T} \right]. \quad (115)$$

For the RKS gas, the excluded-volume transformation gives

$$n^{\text{id}} = \frac{n}{1 - bn}. \quad (116)$$

Thus, in the limit $T \rightarrow 0$, a finite particle number density n_0 requires

$$\mu^* \simeq m - \frac{3T}{2} \ln \left(\frac{T}{c_0^{\text{RKS}}} \right), \quad c_0^{\text{RKS}} = \frac{2\pi}{m} \left[\frac{n_0}{d(1 - bn_0)} \right]^{2/3}. \quad (117)$$

Since

$$\mu^* = \mu - \frac{bnT}{1 - bn} + \frac{a}{b} \ln(1 + bn) + \frac{an}{1 + bn}, \quad (118)$$

the physical chemical potential behaves as

$$\mu \simeq m - \frac{a}{b} \ln(1 + bn_0) - \frac{an_0}{1 + bn_0} - \frac{3T}{2} \ln \left(\frac{T}{c_0^{\text{RKS}}} \right). \quad (119)$$

Therefore, the entropy density of the Boltzmann RKS gas at $T \rightarrow 0$ is

$$s_{\text{RKS}}^{\text{Boltz}} \simeq n_0 \left[\frac{5}{2} + \frac{3}{2} \ln \left(\frac{T}{c_0^{\text{RKS}}} \right) \right]. \quad (120)$$

It becomes negative in the zero-temperature limit, in contradiction with the third law of thermodynamics. Therefore, quantum statistics is needed to describe a physical system at $T \rightarrow 0$.

3 RKS Equation of State with Quantum Statistics

Quantum generalization of the RKS equation of state should satisfy the following requirements:

1. It should reduce to the ideal quantum gas when $a = 0$ and $b = 0$.
2. It should reduce to the classical RKS equation of state in the thermodynamic region where quantum statistics can be neglected.
3. The entropy density should be non-negative and should go to zero when $T \rightarrow 0$.

The pressure of the ideal quantum gas in the GCE reads

$$p^{\text{id}}(T, \mu) = \frac{d}{3} \int \frac{d^3k}{(2\pi)^3} \frac{k^2}{\sqrt{m^2 + k^2}} \left[\exp \left(\frac{\sqrt{m^2 + k^2} - \mu}{T} \right) + \eta \right]^{-1}, \quad (121)$$

where $\eta = +1$ for Fermi statistics, $\eta = -1$ for Bose statistics, and $\eta = 0$ for the Boltzmann approximation.

The RKS pressure in the quantum-statistical formulation is written as

$$p(T, \mu) = p^{\text{id}}(T, \mu^*) - \frac{an^2}{1 + bn}, \quad (122)$$

where the shifted chemical potential is

$$\mu^* = \mu - b p^{\text{id}}(T, \mu^*) + \frac{a}{b} \ln(1 + bn) + \frac{an}{1 + bn}. \quad (123)$$

The particle number density is obtained from

$$n(T, \mu) = \frac{n^{\text{id}}(T, \mu^*)}{1 + b n^{\text{id}}(T, \mu^*)}, \quad (124)$$

where

$$n^{\text{id}}(T, \mu) = \frac{d}{2\pi^2} \int_0^\infty dk k^2 \left[\exp \left(\frac{\sqrt{m^2 + k^2} - \mu}{T} \right) + \eta \right]^{-1}. \quad (125)$$

The entropy density has the form

$$s(T, \mu) = \frac{s^{\text{id}}(T, \mu^*)}{1 + b n^{\text{id}}(T, \mu^*)}, \quad (126)$$

and is always non-negative for the quantum ideal gas. Therefore, the quantum RKS equation of state satisfies the third law of thermodynamics.

Assuming that a and b are temperature independent, the energy density can be calculated as

$$\epsilon(T, \mu) = \frac{\epsilon^{\text{id}}(T, \mu^*)}{1 + b n^{\text{id}}(T, \mu^*)} - \frac{an}{b} \ln(1 + bn). \quad (127)$$

Equivalently, one can rewrite the RKS pressure as a function of temperature T and particle number density n . From

$$n^{\text{id}}(T, \mu^*) = \frac{n}{1 - bn}, \quad (128)$$

one obtains

$$\mu^*(n, T) = \mu^{\text{id}}\left(\frac{n}{1 - bn}, T\right), \quad (129)$$

where $\mu^{\text{id}}(n, T)$ is the ideal quantum-gas chemical potential.

Therefore, the RKS equation of state with quantum statistics can be written as

$$p(T, n) = p^{\text{id}}\left[T, \mu^{\text{id}}\left(\frac{n}{1 - bn}, T\right)\right] - \frac{an^2}{1 + bn}. \quad (130)$$

In the Boltzmann limit, one has

$$p^{\text{id}}(T, \mu^*) = T n^{\text{id}}(T, \mu^*), \quad (131)$$

and Eq. (130) reduces to the classical RKS equation of state,

$$p(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + bn}. \quad (132)$$

4 Quantum van der Waals theory meets quarkyonic matter

Mixture of Nucleons and Quarks with Pauli Exclusion Principle

The densities of baryon number, electric charge, and energy contain contributions from quarks and baryons and read

$$\begin{aligned} \rho_B &= n_p^{\text{id}}(k_F^p) + n_n^{\text{id}}(k_F^n) + n_u^{\text{id}}(q_F^u) + n_d^{\text{id}}(q_F^d), \\ \rho_Q &= n_p^{\text{id}}(k_F^p) + 2n_u^{\text{id}}(q_F^u) - n_d^{\text{id}}(q_F^d), \\ \epsilon &= \epsilon_p^{\text{id}}(k_F^p) + \epsilon_n^{\text{id}}(k_F^n) + \epsilon_u^{\text{id}}(q_F^u) + \epsilon_d^{\text{id}}(q_F^d). \end{aligned}$$

After formalte the pauli principle we have that:

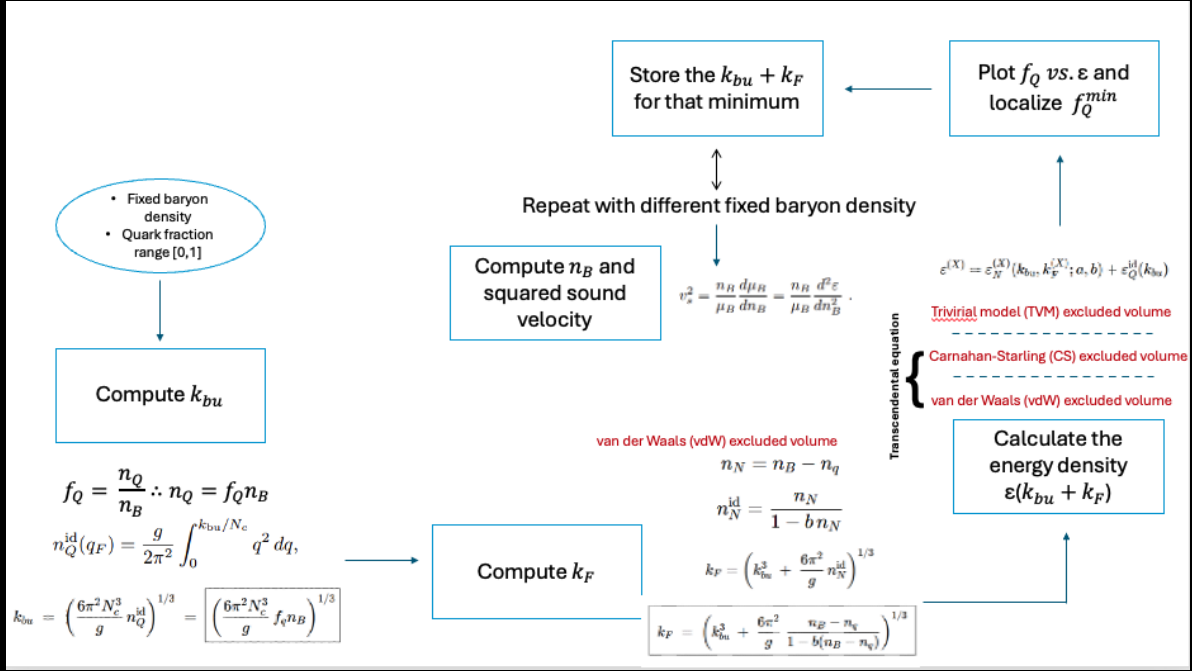
$$\begin{aligned} n_u &= \frac{1+y}{2-y} \frac{g}{2\pi^2} \int_0^{h_{b,u}/N_c} q \sqrt{q^2 + \Lambda^2} dq, \\ n_d &= \frac{g}{2\pi^2} \int_0^{h_{b,u}/N_c} q \sqrt{q^2 + \Lambda^2} dq. \end{aligned}$$

The ideal gas functions are given as

$$\begin{aligned} n_{p,n}^{\text{id}}(k_F^{p,n}) &= \frac{g}{2\pi^2} \int_0^{k_F^{p,n}} k^2 \rho_{p,n}(k) dk, \\ \epsilon_{p,n}^{\text{id}}(k_F^{p,n}) &= \frac{g}{2\pi^2} \int_0^{k_F^{p,n}} k^2 \rho_{p,n}(k) \epsilon_{p,n}(k) dk, \\ n_{u,d}^{\text{id}}(q_F^{u,d}) &= \frac{g}{2\pi^2} \int_0^{q_F^{u,d}} q^2 \rho_{u,d}(q) dq, \\ \epsilon_{u,d}^{\text{id}}(q_F^{u,d}) &= N_c \frac{g}{2\pi^2} \int_0^{q_F^{u,d}} q^2 \rho_{u,d}(q) \epsilon_{u,d}(q) dq. \end{aligned}$$

Meanwhile for the ideal gas we got the fermi momentum expressed as:

$$n_{p,n}^{\text{id}} = \frac{g}{2\pi^2} \int_{k_{bu}}^{k_F^{p,n}} k^2 dk = \frac{g}{6\pi^2} \left[(k_F^{p,n})^3 - k_{bu}^3 \right].$$



5 Quantum van der Waals quarkyonic matter at nonzero isospin asymmetry

6 Correlations between nuclear incompressibility, liquid-gas critical point, and quarkyonic transition

6.1 Equations of State of Real Gases

Classical equations of state for real gases are usually presented in terms of the pressure P as a function of temperature T and particle number density $n \equiv N/V$. In Ref. [?] several examples of real gas models were used to calculate the properties of nuclear matter:

- van der Waals (vdW)

$$P_{vdW}(T, n) = \frac{nT}{1 - bn} - an^2, \quad (133)$$

- Redlich-Kwong-Soave (RKS)

$$P_{RKS}(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + bn}, \quad (134)$$

- Peng-Robinson (PR)

$$P_{PR}(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{1 + 2bn - (bn)^2}, \quad (135)$$

- and Clausius

$$P_C(T, n) = \frac{nT}{1 - bn} - \frac{an^2}{(1 + cn)^2}. \quad (136)$$

The first term in Eqs. (133)–(136) include repulsive interactions in the framework of excluded volume. Here the parameter b regulates the strength of these interactions. The second term in Eqs. (133)–(136) describes the different forms of mean-field attraction regulated by the parameter a . Note that Eq. (136) includes also a third parameter $c \geq 0$, which additionally regulates attractive interactions, suppressing them with increasing c . At $c = 0$ the Clausius model is reduced to the vdW model.

In addition to the above equations for real gases we consider the Dieterici EoS. For our purposes we formulate the generalized Dieterici equation in the following form ($5/3 \leq \alpha \leq 2$):

$$P_D(T, n) = \frac{nT}{1 - bn} - an^\alpha. \quad (137)$$

Dieterici originally proposed this equation in 1899 [?] with $\alpha = 5/3$. Here we use Dieterici equation in a generalized form with arbitrary α parameter which allows a smooth transition from the original Dieterici model ($\alpha = 5/3$) to the vdW model ($\alpha = 2$).

6.2 Ground state of nuclear matter

The model parameters are constrained by nuclear ground state properties. In the nuclear ground state $T = 0$ and the total energy has a minimum (correspondingly $P = 0$) with the following values of nucleon number density n_0 and binding energy per nucleon W_0 :

$$n_0 = 0.16 \text{ fm}^{-3}, \quad (138)$$

$$W_0 \equiv \frac{\epsilon(T=0, n=n_0)}{n_0} - m = -16 \text{ MeV} \quad (139)$$

For the GCE, we have the transcendental equations:

$$P(T, \mu) = p^{id}(T, \mu^*) + n^2 u'(n), \quad (140)$$

$$\epsilon(T, \mu) = (1 - bn)\epsilon^{id}(T, \mu^*) + nu(n) \quad (141)$$

$$n(T, \mu) = (1 - bn)n^{id}(T, \mu^*) \quad (142)$$

$$\mu^* = \mu - bp^{id}(T, \mu^*) - u(n) - nu'(n) \quad (143)$$

where μ is the chemical potential, μ^* is the effective chemical potential, ϵ is the energy density, and the function $u(n)$ corresponds to the energy of attractive interactions per particle.

and for vdW specifically:

$$u(n) = -an, \quad (144)$$

therefore

$$u'(n) = -a. \quad (145)$$

Then:

$$P(T, \mu) = p^{id}(T, \mu^*) - an^2, \quad (146)$$

$$\epsilon(T, \mu) = (1 - bn)\epsilon^{id}(T, \mu^*) - an^2. \quad (147)$$

From equation 142 at $T = 0$ and $n = n_0$

$$n_0 = (1 - bn_0)n^{id}(0, \mu^*) \quad (148)$$

So

$$n^{id}(0, \mu^*) = \frac{n_0}{1 - bn_0} \quad (149)$$

On the other hand we have that the ideal fermi gas functions are defined as:

$$p^{id}(T, \mu) = \frac{d}{6\pi^2} \int_0^\infty dk \frac{k^4}{\sqrt{m^2 + k^2}} \left[\exp\left(\frac{\sqrt{m^2 + k^2} - \mu}{T}\right) + 1 \right]^{-1},$$

$$n^{id}(T, \mu) = \left(\frac{\partial p^{id}}{\partial \mu} \right)_T,$$

$$\epsilon^{id}(T, \mu) = T \left(\frac{\partial p^{id}}{\partial T} \right)_\mu + \mu n^{id} - p^{id}.$$

Algorithm

1. Set the physical constants:

$$n_0 = 0.16 \text{ fm}^{-3}, \quad m \simeq 938 \text{ MeV}, \quad d = 4. \quad (165)$$

2. Choose a trial value of b .

3. Compute the effective ideal-gas density:

$$n_* = \frac{n_0}{1 - bn_0}. \quad (166)$$

4. Compute the corresponding Fermi momentum and Fermi energy:

$$k_F^* = \left(\frac{6\pi^2 n_*}{d} \right)^{1/3}, \quad E_F^* = \sqrt{(k_F^*)^2 + m^2}. \quad (167)$$

5. Evaluate the ideal Fermi-gas functions:

$$\varepsilon^{\text{id}}(b), \quad p^{\text{id}}(b). \quad (168)$$

6. Build the scalar function

$$f(b) = p^{\text{id}}(b) - (1 - bn_0)\varepsilon^{\text{id}}(b) + n_0(m - 16). \quad (169)$$

7. Solve numerically

$$f(b) = 0 \quad (170)$$

8. Once b is known, compute

$$a = \frac{p^{\text{id}}(b)}{n_0^2}. \quad (171)$$

9. Construct the full zero-temperature pressure as a function of density:

$$P(0, n) = p^{\text{id}}\left(\frac{n}{1 - bn}\right) - an^2. \quad (172)$$

10. Compute the derivative of the pressure at saturation:

$$\left(\frac{\partial P}{\partial n} \right)_{T=0, n=n_0}. \quad (173)$$

In practice, this is usually done numerically by a central difference:

$$\left(\frac{\partial P}{\partial n} \right)_{n_0} \approx \frac{P(0, n_0 + h) - P(0, n_0 - h)}{2h}. \quad (174)$$

11. Finally evaluate the incompressibility:

$$K_0 = 9 \left(\frac{\partial P}{\partial n} \right)_{T=0, n=n_0}. \quad (175)$$

Input: n_0, m, d
 Solve $f(b) = 0$,
 $f(b) = p^{\text{id}}(b) - (1 - bn_0)\varepsilon^{\text{id}}(b) + n_0(m - 16)$,
 $a = \frac{p^{\text{id}}(b)}{n_0^2}$,
 $K_0 = 9 \left(\frac{\partial P}{\partial n} \right)_{T=0, n_0}$.

(176)

From equation for 142 at ground state ($T = 0, P = 0, n = n_0 = -0.16 \text{ fm}^{-3}, W_0 = -16 \text{ MeV}$)

$$n(0, \mu) = (1 - bn(0, \mu))n^{id}(0, \mu^*) = n_0 = (1 - bn_0)n^{id}(0, \mu^*) \quad (177)$$

solving for n^{id} :

$$\frac{n_0}{(1 - bn_0)} = n^{id}, \quad (178)$$

now we can fix b and get n^{id} . On the other hand for pressure 140, we have:

$$P(0, \mu) = p^{id} + n^2 u'(n) = 0, \quad (179)$$

therefore solving for p^{id}

$$-n_0^2 u'(n) = p^{id}, \quad (180)$$

and $u'(n) \equiv u'(n, a, b)$, so with fix b we can numerically get a , but we still need to know p^{id} which from eq 6.2 we have at $T = 0$,

$$p^{id}(0, \mu) = \frac{d}{6\pi^2} \int_0^{k_f} \frac{k^2}{\sqrt{k^2 + m^2}} \quad (181)$$

this can be computed computationally knowing that at ground state, for nuclear symmetric matter:

$$k_f = \left(\frac{6\pi^2 n^{id}}{d} \right)^{1/3} \quad (182)$$

with $d = 4$ and eq 178 we have:

$$k_f = \left(\frac{6\pi^2}{4} \frac{n_0}{(1 - bn_0)} \right)^{1/3} \quad (183)$$

So after integrating we can plug into the result.

To determine the parameter b , we impose the ground-state condition for symmetric nuclear matter at $T = 0$,

$$\frac{\varepsilon}{n_0} - m = -16 \text{ MeV}, \quad (184)$$

with

$$\varepsilon(0, \mu) = (1 - bn_0) \varepsilon^{id}(0, \mu^*) + n_0 u(n_0). \quad (185)$$

Using also

$$n_0 = (1 - bn_0) n^{id}(0, \mu^*), \quad (186)$$

we divide by n_0 and obtain

$$\frac{\varepsilon}{n_0} = \frac{(1 - bn_0) \varepsilon^{id}(0, \mu^*)}{n_0} + u(n_0) = \frac{\varepsilon^{id}(0, \mu^*)}{n^{id}(0, \mu^*)} + u(n_0). \quad (187)$$

Therefore, the binding-energy condition becomes

$$\frac{\varepsilon^{id}(0, \mu^*)}{n^{id}(0, \mu^*)} + u(n_0) - m = -16 \text{ MeV}. \quad (188)$$

Note that ε^{id} is the ideal Fermi-gas energy density, not the chemical potential. At $T = 0$, it is obtained from the Fermi integral

$$\varepsilon^{id}(n^*) = \frac{d}{2\pi^2} \int_0^{k_F} k^2 \sqrt{k^2 + m^2} dk, \quad (189)$$

where k_F is

$$k_F = \left(\frac{6\pi^2}{d} \frac{n_0}{1 - bn_0} \right)^{1/3}. \quad (190)$$

Thus, for each trial value of b , one computes n^{id} , then k_F , and evaluates $\varepsilon^{id}(n^{id})$. The correct value of b is the one that satisfies

$$\frac{\varepsilon^{id}(n^*)}{n^*} + u(n_0) - m = -16 \text{ MeV}, \quad (191)$$

together with the zero-pressure condition

$$p^{id}(n^*) + n_0^2 u'(n_0) = 0. \quad (192)$$

6.3 Computing critical points

The location of the CP, T_c and n_c is defined by the following set of eqs.:

$$\left(\frac{\partial P}{\partial n}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial n^2}\right)_T = 0 \quad (193)$$

6.3.1 Van der Waals example (classical)

From eq. 133

$$P_{\text{vdW}}(T, n) = \frac{nT}{1 - bn} - an^2, \quad (194)$$

computing first derivative:

$$\left(\frac{\partial P}{\partial n}\right)_T = \frac{T}{1 - bn} + \frac{nT}{(1 - bn)^2}(-b) - 2an = 0 \quad (195)$$

$$\frac{T(1 - bn) + bnT}{(1 - bn)^2} = 2an \quad (196)$$

$$\frac{T - Tbn + bnT}{(1 - bn)^2} = 2an \quad (197)$$

$$\frac{T}{(1 - bn)^2} = 2an \quad (198)$$

Then the second derivative is:

$$\frac{\partial}{\partial n} \left[\frac{T}{(1 - bn)^2} \right] = \frac{\partial}{\partial n} [2an] \quad (199)$$

$$\frac{T}{(1 - bn)^3}(-2)(-b) = 2a \quad (200)$$

$$\frac{2bT}{(1 - bn)^3} = 2a \quad (201)$$

Calculating the ratio between both:

$$\frac{\frac{2bT}{(1 - bn)^3}}{\frac{T}{(1 - bn)^2}} = \frac{2a}{an} \quad (202)$$

Now solving n_c from here:

$$\frac{2b}{(1 - bn_c)} = \frac{1}{n_c} \quad (203)$$

$$2bn_c = 1 - bn_c \rightarrow 3bn_c = 1 \quad (204)$$

$$n_c = \frac{1}{3b} \quad (205)$$

Now, substituting this value into either of the derivatives and solving for T_c one obtains:

$$T_c = \frac{8a}{27b} \quad (206)$$

However we are currently in the GCE and it is not classical anymore as seen in the first section of this document, so we need to find T_c and n_c from the actual transcendental equations 140

At $T > 0$, the ideal density and pressure are

$$n^{\text{id}}(T, \mu^*) = \frac{d}{2\pi^2} \int_0^\infty k^2 \frac{dk}{\exp\left[\frac{E(k) - \mu^*}{T}\right] + 1}, \quad (207)$$

$$p^{\text{id}}(T, \mu^*) = \frac{d}{6\pi^2} \int_0^\infty \frac{(\hbar c)^2 k^4}{E(k)} \frac{dk}{\exp\left[\frac{E(k) - \mu^*}{T}\right] + 1}. \quad (208)$$

Then the physical density is obtained through the excluded-volume relation

$$n = (1 - bn) n^{\text{id}}(T, \mu^*), \quad (209)$$

or equivalently,

$$n = \frac{n^{\text{id}}(T, \mu^*)}{1 + b n^{\text{id}}(T, \mu^*)}. \quad (210)$$

Finally, the interacting pressure is

$$P(T, \mu) = p^{\text{id}}(T, \mu^*) + a n^2 u'(n). \quad (211)$$

Thus, for each fixed temperature T , one can generate the isotherm $P = P(n, T)$ parametrically by varying μ^* :

$$\mu^* \longrightarrow n^{\text{id}}(T, \mu^*), \quad p^{\text{id}}(T, \mu^*) \longrightarrow n(T, \mu^*) \longrightarrow P(T, \mu^*). \quad (212)$$

6.3.2 8. Physics of the critical point

The critical point is the endpoint of the liquid–gas first-order phase transition.

- For $T < T_c$, the isotherm has a liquid–gas structure: two phases can coexist.
- For $T = T_c$, the coexistence region shrinks to one point.
- For $T > T_c$, there is no first-order transition anymore, only smooth crossover behavior.

In the P -vs- n plane, the critical isotherm has a horizontal inflection point. Therefore,

$$\left(\frac{\partial P}{\partial n}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial n^2}\right)_T = 0 \quad \text{at} \quad (T, n) = (T_c, n_c). \quad (213)$$

This is the precise mathematical definition of the critical point.

6.3.3 9. Mathematical procedure to compute T_c and n_c

Once a and b are already fixed from the ground state, the finite-temperature calculation is:

1. Choose a trial temperature T .
2. Vary μ^* over a grid.
3. For each μ^* , compute $n^{\text{id}}(T, \mu^*)$ from Eq. (207).
4. Compute $p^{\text{id}}(T, \mu^*)$ from Eq. (208).
5. Convert n^{id} to the physical density n using Eq. (209).
6. Compute the interacting pressure P using Eq. (211).
7. In this way, build the full isotherm $P(n, T)$.

8. Differentiate the isotherm with respect to n .

9. Find the point where

$$\left(\frac{\partial P}{\partial n}\right)_T \approx 0, \quad \left(\frac{\partial^2 P}{\partial n^2}\right)_T \approx 0.$$

10. Repeat this for many temperatures.

11. The temperature where both derivatives vanish simultaneously is T_c , and the corresponding density is n_c .

So the logic is:

$$(a, b) \text{ fixed from ground state} \implies P(n, T) \text{ for many } T \implies (T_c, n_c). \quad (214)$$

6.3.4 10. Compact summary

The whole method can be summarized as follows:

$$\text{Ground state:} \quad T = 0, \quad P = 0, \quad n = n_0, \quad \frac{\varepsilon}{n_0} - m = -16 \text{ MeV}, \quad (215)$$

$$\text{Step 1:} \quad n^{\text{id}} = \frac{n_0}{1 - bn_0}, \quad (216)$$

$$\text{Step 2:} \quad k_F = \left(\frac{6\pi^2 n^{\text{id}}}{d}\right)^{1/3}, \quad (217)$$

$$\text{Step 3:} \quad a = -\frac{p^{\text{id}}}{n_0^2 u'(n_0)}, \quad (218)$$

$$\text{Step 4:} \quad \frac{\varepsilon^{\text{id}}}{n^{\text{id}}} + a u(n_0) - m = -16 \text{ MeV} \implies b, \quad (219)$$

$$\text{Step 5:} \quad \text{with } a, b \text{ fixed, compute finite-}T \text{ isotherms,} \quad (220)$$

$$\text{Step 6:} \quad \left(\frac{\partial P}{\partial n}\right)_T = 0, \quad \left(\frac{\partial^2 P}{\partial n^2}\right)_T = 0 \implies (T_c, n_c). \quad (221)$$

Asymmetric Quarkyonic Nuclear Matter

$$v(n_j) = - \sum_{i,j \in B} \frac{a_{ij} n_j}{1 + c n_B}$$